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Usable computing systems

Creating a usable website by considering human-computer interaction

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Abstract

It is vital for websites to be usable and accessible in order for them to be successful, especially if they are the main marketing front for a company. User Experience should be a key consideration when designing a website in order to ensure that it has those qualities. A wide variety of tools are available in order to determine, during the design phase of building a website, whether these have been appropriately implemented.

User Engagement should also be considered, though research regarding this is more limited and tools to measure it seem limited if not non-existent at this current point in time. Additionally, all areas of website design should be developed with the larger context of Human-Computer Interaction in mind, and the appropriate standards and conventions should always be followed.

1 Introduction

Although a more visually appealing website can lead to greater trust in its company [1], accessibility and usability are far more important when building a successful website for a business. It is especially important to consider this in the context of Human-Computer Interaction (HCI), which Hewett et al. define as "a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them" [2]. It is the former of these that we are concerned with in this report.

Newer companies with fewer workers can struggle to produce effective and usable websites, resulting in said websites becoming difficult and counter-intuitive to navigate. This can be especially harmful to said companies if their website doubles as their primary sales tool, as is the case for companies such as Find My Gig, because prospective customers will instead leave in favour of a more established company whose website is easier to use. This can cause a drop in revenue of up to 50% compared to companies with a well-designed website [3]. As a result, fixing these issues can become a key priority for smaller companies, but due to their size, they can struggle to determine what is necessary in order to do so. This report aims to tackle that dilemma.

This report will comprise of two primary sections. The first one of these discusses how to build a usable website, including what the top considerations in doing so should be, with the objective of improving user retention and consequently sales. The second section will outline how accessibility and ease of use play a role in usable websites, as well as explain how to ensure these are present while building one. Additionally, this report will cover modern academic literature discussing HCI, especially including research and focusing on the past two calendar years. It will also cover standards and conventions that should be followed in order to create usable websites.

2 User Experience

User Experience (UX), which David Benyon [4] describes as how people interact with something and the impact this has on them, is important for designing a good web page. If the impact something has on people is predominantly negative, for example, it is less likely that it will be interacted with in the future. Benyon further

expands this, stating that those involved in UX should consider every situation the user may be in while interacting with the product being created, providing the example of the iPhone – its touchscreen might be good UX in most scenarios, but on a cold winter's day, the user may be wearing gloves, and consequentially the UX is much poorer in that scenario.

Additionally, UX is important in the context of Human-Computer Interaction (HCI). This is something that has started gaining traction in recent years, replacing usability as the dominant concept among researchers and companies in the "HCI community" [5], and there are many systems used for measuring usability and UX with regards to HCI. The most popular of these is the System Usability Scale, a "simple, ten-item scale giving a global view of subjective assessments of usability" [6], though a modern competitor is emerging in the form of the User Experience Questionnaire, a "fast and direct measurement of UX" that was "designed for use as part of a normal usability test, but also as an online questionnaire" [7] and is available (as of writing) at https://www.ueq-online.org/. This is both a symptom and an indicator of the shift towards UX by those involved in HCI.

3 Analytics

Every factor of a website's performance should be monitored, even once the website has been released to the public, and UX is no exception to this fact. One method to do so with regards to UX is through web analytics and metrics, which have become vital elements in many commercial industries [8]. Fundingsland Jr. et al. point to efficiency, user satisfaction and effectiveness as objectives that can be interpreted as key metrics, which in turn can be broken down into simpler metrics that can be easily tracked using external services such as Google Analytics. They also state that web analytics has become a commercial industry in and of itself, which results in the prospect of companies developing their own services for web analytics not only unfeasible, but also impractical.

Palomino et al. [9] note that the information that is provided to companies by using Web Analytics represents "a considerable statistical sample of users", which allows for a significantly comprehensive analysis of statistics, trends and, consequentially, user experiences. This is done through analysing log files in order to determine a user's behaviour, both "on-site" (information collected from activity on the website, such as average duration, common activities - or lack of such - or number of visits) and "off-site" (information collected from elsewhere, including surveys, information regarding similar and/or competing websites, and other, generally external, sources). They also determine that Google Analytics is, by a significant margin, the most used Web Analytics tool, reporting that this is due to its usefulness regarding both tracking and reporting on-site data, as well as its multiple functionalities, ease of use and free online access.

However, there are other tools one could use, either as opposed to Google Analytics or in conjunction to it. These include Matomo Analytics, which is proficient in flexibility and speed [10], and CrazyEgg, which focuses on click-through analysis [9]. Another way to gather analytics is to use one, or multiple, of many social media Application Program Interfaces (APIs) [11], which are defined by Robillard et al. as "*the interface to a reusable software entity used by multiple clients outside the developing organization,*

and that can be distributed separately from environment code" [12]. Despite only being useful regarding off-site data, APIs can be used to understand the personal opinions of those using the website, especially beliefs raised inside of complaints. Sharma et al. [11] also mention the existence of the paid tool Hootsuite as an alternative to social media APIs, but do not elaborate on whether its cost is worth any benefits it may have.

4 User Engagement

While it is extremely important, UX should not be the sole consideration when creating a website. A different idea to factor in when doing so is User Engagement, which, as some social media websites have shown [13], is a similar but distinct concept when compared to UX. User Engagement describes the broad experience a user has with something over time (for instance, reasons why they would or would not to return to a product), whereas UX describes specific in-the-moment interactions (for example, whether one enjoys using a product). While the quality of one does not ensure the quality of the other, they are closely linked and amplify each other's effectiveness, as User Engagement considers how the product allows its users to gain value from its UX. Lalmas et al. also explain that User Engagement can be measured in multiple ways, including the previously mentioned analytics. Using this allows information regarding both it and UX to be collected simultaneously, despite the depth of information collected by analytics being relatively shallow when compared to other methods.

It has also been found that making alterations to a website is not the only way to improve user engagement. Another way is to use branded search traffic, which is where advertisements on other locations (primarily other websites) entice potential customers and encourage them to use a "branded search term" in their search engine [14]. This is done in two ways: by redirecting enticed web users (those who click on the advertisement) to the website, which improves the probability of them searching for the website in the future, and by being memorable to other potential customers who do not click on the advertisement, allowing them to potentially search for the website in the future. This promotes the website within the algorithms of search engines (through the "branded search term") while simultaneously improving brand awareness (through the advertisement itself). Additionally, this results in an increased quantity of unique visitors to the website and the average visitor viewing it for an increased duration [15]. When considering the context of a website doubling as a primary sales tool, this is likely to result in increased sales.

5 Accessibility

A significant population of people, including millions of students in India alone [16], are disabled. If a website does not accommodate their issues and needs, they will be more likely to refrain from using it in the future, limiting potential traffic. Disabilities that affect people's experience and ability to interact with computers and related products, such as websites, include impairments to sight (whether blindness, colour blindness or something else) and hearing, poor motor control (such as those suffering from Parkinson's disease), photosensitive epilepsy and mental disabilities (including those on the autistic spectrum).

The World Wide Web Consortium (W3C), who develop and maintain many international web standards such as HTML and CSS, have developed a series of

detailed guidelines through the Web Accessibility Initiative (WAI) [17]. The current relevant version (as of writing) is WCAG 2.2, which was published as a web standard on 5th October 2023 [18]. This standard, in full detail, can be found at https://www.w3.org/TR/WCAG22/.

A comprehensive study of accessibility in both education and industry [19] discovered that design systems were a tool that was growing in popularity in order to ensure accessibility, and was also able to create an "explicit connection" from said tool to accessibility. Despite this, it has been found that HCI has not been significantly considered in the creation of many modern design systems [20], and so one should research carefully in order to ensure that their product's design system does consider HCI and that the accessibility of their product is maximised.

There are multiple tools that can be used to evaluate accessibility of websites. One of these [16] is aXe, an open-source tool that runs as a browser extension. It provides those that use it with a list of issues, including how many times those issues appear, detailed descriptions of them, and information regarding how severe the issues are and why they should be corrected. The tool was developed by Deque, an American company that specialises in digital accessibility [21]. Another tool [16] is TAW ("Test de Accesibilidad Web"), which marks any accessibility issues it finds and provides proposals on how to fix them. Ismail and Kuppusamy highlight both its clarity and its compatibility with WCAG 1.0 and 2.0, though it is worth noting that the website claims to also be compatible with WCAG 2.1 [22]. It was developed by CTIC ("La Fundación Centro Tecnológico de la Información y la Comunicación"), a Spanish non-profit that assists companies in the fields of ICT and computer science [23, 24]. Both of the methods listed are free to use, and TAW is available in Spanish, English and Portuguese.

6 Ease of Use

Despite being replaced by UX as the dominant concept in the "HCI community" [5], usability is still a concept that should be considered in the creation of a website. A product is usable if it is simple, adaptable to its current situation and ensures the happiness of its user [25], and as such websites with poor usability are less likely to receive significant future traffic. Additionally, evaluating usability allows software development teams to determine both the positive and negative attributes of their prototypes, which helps with reflection and improves the efficacy and effectiveness of the final product. Usability is especially necessary alongside HCI with regards to smart and portable devices [26].

There are also many common conventions present on commercial websites, and for websites that follow them, their users find navigation simpler as certain aspects of the websites are in their expected locations. These include [27]:

- The company logo being in the top-left corner of the screen. This is also often a link to the home page.
- A link to the basket, with an appropriate image stating the number of items it contains, is in the top-right corner of the screen.
- A search bar takes up the majority of the top row, being placed between the logo and the basket.
- A horizontal hyperlinked list of main sections, also known as a navigation bar, is present underneath these three.
    - There is often a hamburger menu in addition to this, which

is more effective when labelled to show its meaning (it presents more options when clicked), but this is less common.

- Between the navigation bar and the main content is a breadcrumb trail to allow the user to navigate between content more easily.

Additionally, including user participation in testing will allow for alternative perspectives and highlight issues that had not been discovered or considered prior, increasing the usability of the final end product. This can also be used for accessibility purposes, allowing those that would usually be disadvantaged when using the product to provide feedback.

7 Conclusion

When it comes to designing effective websites, the three primary considerations should be User Experience (UX), accessibility and usability. There are many methods to ensure that these essential design elements are considered, including the User Experience Questionnaire [7], TAW [16] and following accessibility guidelines such as WCAG 2.2 [18] and industry conventions [27]. There are other elements to consider regarding web design in addition to these, however, and though they are less important, they should not be discounted and can even be used to support and develop the primary considerations. These include User Engagement [9], design systems [13] and the overall consideration of HCI.

When taking this advice into account, one must remember that UX is subjective, and that people have differing opinions on what comprises a positive UX. This is true both when considering different website designers, different users, and the differing points of view between users and website designers. As a result, this is a dynamic field, and opinions, standards and conventions with regards to what makes a website usable and accessible change over time. For example, WCAG 3.0 is currently in development, with the next draft expected within the next month (as of writing), though it is not expected to be completed within the next few years [28]. Additionally, the sources, standards and conventions cited are dominated by ideas originating in Europe and the United States of America - different cultures may have alternate views regarding the topics covered. These are all areas that, if researched and discussed in detail as part of future research, would allow for significant improvement and a more global consensus regarding the elements and methods required to create a usable and accessible website.

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